1. Create a splash screen that displays a logo and app name for 3 seconds before launching the main activity.

2. Modify the splash screen to include a fade-in animation for the logo.

5. Add a background image to the splash screen and ensure it scales properly.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppSplashScreen">
        <activity
            android:name=".MainActivitySplash"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".MainActivity"
            android:exported="false">

        </activity>
    </application>

</manifest>
```

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:textSize="30dp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.appsplashscreen;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main_splash.xml

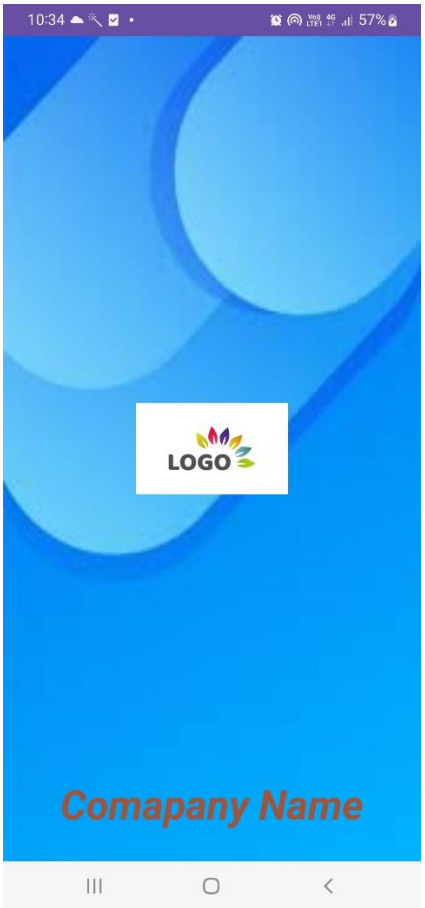
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#DFE2C7"
    tools:context=".MainActivitySplash">

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <ImageView
            android:id="@+id/imageView2"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:scaleType="fitXY"
            app:srcCompat="@drawable/img_1" />
        <ImageView
            android:id="@+id/imageView"
            android:layout_width="150dp"
            android:layout_height="150dp"
            android:layout_gravity="center"
            app:srcCompat="@drawable/img" />
        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="bottom|center_horizontal"
            android:text="Comapany Name"
            android:layout_margin="30dp"
            android:textSize="40dp"
            android:textColor="#A85338"
            android:textStyle="bold|italic"
            />
    </FrameLayout>

</LinearLayout>
```

MainActivitySplash.java

```
package com.example.appsplashscreen;

import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.ImageView;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivitySplash extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_splash);


      //following code is for animation

         ImageView img = findViewById(R.id.imageView);
        Animation fadeInAnimation = AnimationUtils.loadAnimation(this,
R.anim.fade_in);
        img.startAnimation(fadeInAnimation);


        // following code if tostart new main activity after 7 secon
        Handler hd = new Handler(Looper.getMainLooper());
        hd.postDelayed(new Runnable() {
            @Override
            public void run() {
                Intent objIntent = new Intent(MainActivitySplash.this,
MainActivity.class);
                startActivity(objIntent);
                finish();
            }
        }, 7000);
  }
}
```

Res//anim//fade_in.xml file

Above folder and file is newly added for animation

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
   <alpha
      android:fromAlpha="0.0"
      android:toAlpha="1.0"
      android:duration="3000"
      android:interpolator="@android:anim/decelerate_interpolator" />

</set>
```

OUTPUT: